Artificial Intelligence Basics

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Types of AI

1. Narrow AI

Narrow AI is designed to perform a specific task. Examples include voice assistants, recommendation systems, and chatbots.

2. General AI

General AI refers to machines that can perform any intellectual task that a human can do. This type of AI does not yet fully exist.

Machine Learning

Machine Learning is a subset of AI where computers learn patterns from data instead of being explicitly programmed.

Supervised Learning

In supervised learning, models are trained using labeled data.

Unsupervised Learning

In unsupervised learning, models identify patterns from unlabeled data.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with many layers. It is commonly used in image recognition and natural language processing.

Natural Language Processing

Natural Language Processing (NLP) helps computers understand and generate human language.

Examples include:

- ChatGPT
- Translation systems
- Voice assistants

Retrieval-Augmented Generation (RAG)

RAG combines retrieval systems with large language models.

Steps in RAG:

1. User asks a question
2. Relevant documents are retrieved
3. Context is sent to the language model
4. Model generates an answer

RAG helps reduce hallucinations by grounding answers in real documents.

Hallucination in AI

Hallucination occurs when an AI model generates false or misleading information.

Methods to reduce hallucination:

- Using RAG
- Fact-checking
- Confidence scoring
- Self-correction loops

Vector Databases

Vector databases store embeddings generated from text.

Popular vector databases:

- ChromaDB
- Pinecone
- FAISS

Ollama

Ollama allows users to run large language models locally on their computers.

Popular Ollama models:

- Llama3
- Mistral
- Phi

Streamlit

Streamlit is a Python framework used to build interactive AI web applications quickly.